A Living Review of Quantum Information Science in High Energy Physics

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ABSTRACT: Inspired by "A Living Review of Machine Learning for Particle Physics"[1], the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

[1] See https://github.com/iml-wg/HEPML-LivingReview.

Contents

Contents

1 Introduction

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. The papers are listed in reverse chronological order. In order to be as useful as possible, this document will continually change. Please check back [2] regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

2 Nuclear and Particle Physics in Quantum Information Science

2.1 Reviews and Whitepapers

2.1.1 Reviews

2.1.1.1 Quantum Machine Learning in High Energy Physics [1]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

[2] See https://github.com/PamelaPajarillo/NUPAQIS-LivingReview.

2.1.2 Whitepapers

2.1.2.1 Mechanical quantum sensing in the search for dark matter [2]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020

- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.1.2.2 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [3]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma

- **Posted on arXiv:** 09 March 2022

2.2 Anomaly Detection

2.2.1 Quantum k-Means

2.2.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [4]

- **Authors:** Andrew Blance, Michael Spannowsky

- **Posted on arXiv:** 05 March 2021

- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.2.2 Variational Quantum Algorithms

2.2.2.1 Anomaly detection in high-energy physics using a quantum autoencoder [5]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi

- **Posted on arXiv:** 09 December 2021

- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

2.2.3 Crosslists

2.2.3.1 A Quantum Algorithm for Model-Independent Searches for New Physics [6]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.3 Beyond the Standard Model - Heavy Vector Boson Searches

2.3.1 Crosslists

2.3.1.1 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [7]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.4 Beyond the Standard Model - Dark Sector Searches

2.4.1 Quantum Algorithms - Grover's Search Algorithm

2.4.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [8]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

2.5 Beyond the Standard Model - Supersymmetry

2.5.1 Crosslists

2.5.1.1 Event Classification with Quantum Machine Learning in High-Energy Physics [9]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.5.1.2 Quantum algorithm for the classification of supersymmetric top quark events [10]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.6 Dark Matter - Axions and Axion-like Particles

2.6.1 Quantum Sensors - Nitrogen-Vacancy Centers

2.6.1.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [11]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du
- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

2.6.1.2 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [12]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

2.6.2 Quantum Sensors - Superconducting Transmon Qubits

2.6.2.1 Accelerating dark-matter axion searches with quantum measurement technology [13]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

2.6.3 Crosslists

2.6.3.1 Mechanical quantum sensing in the search for dark matter [2]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao

- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

2.7 Detector Technology and Detector Simulation

2.7.1 Continuous Variable Quantum Computing

2.7.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [14]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

2.7.2 Quantum Generative Adversarial Networks

2.7.2.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [15]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

2.8 Dark Matter - Exotic Dark Matter

2.8.1 Quantum Sensors - Nitrogen-Vacancy Centers

2.8.1.1 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [16]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

2.9 Dark Matter - Hidden/Dark Photons

2.9.1 Quantum Sensors - Superconducting Transmon Qubits

2.9.1.1 Searching for Dark Matter with a Superconducting Qubit [17]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

2.10 Dark Matter - Large Composite States

2.10.1 Quantum Sensors - Levitated Sensors

2.10.1.1 Search for composite dark matter with optically levitated sensors [18]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

2.11 Dark Matter - Low Mass Dark Matter

2.11.1 Quantum Sensors - Levitated Sensors

2.11.1.1 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [19]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

2.12 Dark Matter - Millicharged Particles

2.12.1 Quantum Sensors - Trapped Ions

2.12.1.1 Trapped Electrons and Ions as Particle Detectors [20]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

2.12.1.2 Millicharged Dark Matter Detection with Ion Traps [21]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

2.13 Dark Matter - Weakly Interacting Massive Particles

2.14 Effective Field Theories

2.14.1 Quantum Simulations

2.14.1.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [22]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

2.14.2 Crosslists

2.14.2.1 Quantum SMEFT tomography: Top quark pair production at the LHC [23]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.15 Event Classification

2.15.1 Quantum Algorithms - Grover's Search Algorithm

2.15.1.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [24]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.15.2 Quantum Annealing

2.15.2.1 Solving a Higgs optimization problem with quantum annealing for machine learning [25]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

2.15.2.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [26]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

2.15.2.3 Quantum algorithm for the classification of supersymmetric top quark events [10]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

2.15.2.4 Completely quantum neural networks [27]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

2.15.3 Quantum Kernel Methods

2.15.3.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [28]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

2.15.3.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [29]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.15.3.3 Higgs analysis with quantum classifiers [30]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.15.4 Variational Quantum Algorithms

2.15.4.1 Event Classification with Quantum Machine Learning in High-Energy Physics [9]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

2.15.4.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [7]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

2.15.4.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [31]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.15.5 Crosslists

2.15.5.1 Quantum Machine Learning in High Energy Physics [1]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

2.15.5.2 Unsupervised event classification with graphs on classical and photonic quantum computers [4]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

2.16 Event Generation

2.16.1 Quantum Algorithms - Grover's Search Algorithm

2.16.1.1 Quantum algorithm for Feynman loop integrals [32]

- **Authors:** Selomit Ramírez-Uribe, Andrès E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

2.16.2 Quantum Generative Adversarial Networks

2.16.2.1 Style-based quantum generative adversarial networks for Monte Carlo events [33]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

2.16.2.2 Quantum integration of elementary particle processes [34]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

2.16.3 Quantum Simulations

2.16.3.1 Quantum Algorithm for High Energy Physics Simulations [35]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

2.16.3.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [36]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

2.16.4 Quantum Walks

2.16.4.1 Quantum walk approach to simulating parton showers [37]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

2.16.5 Variational Quantum Algorithms

2.16.5.1 Partonic collinear structure by quantum computing [38]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.16.5.2 Unsupervised quantum circuit learning in high energy physics [39]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

2.17 Gravitation and Astrophysics

2.17.1 Quantum Annealing

2.17.1.1 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [40]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

2.17.2 Crosslists

2.17.2.1 Search for composite dark matter with optically levitated sensors [18]

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2.17.2.2 Searching for Dark Matter with a Superconducting Qubit [17]

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2.18 Higgs Physics

2.18.1 Crosslists

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- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

2.18.1.4 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [29]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei

- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

2.18.1.5 Higgs analysis with quantum classifiers [30]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

2.18.1.6 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [24]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

2.19 Jet Reconstruction

2.19.1 Quantum Algorithms - Grover's Search Algorithm

2.19.1.1 Quantum Algorithms for Jet Clustering [41]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

2.19.2 Quantum Annealing

2.19.2.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [42]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

2.19.2.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [43]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

2.19.3 Quantum k-Means

2.19.3.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [44]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

2.19.4 Tensor Networks

2.19.4.1 Quantum-inspired machine learning on high-energy physics data [45]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

2.19.4.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [46]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

2.19.4.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [47]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

2.19.5 Variational Quantum Algorithms

2.19.5.1 Quantum Machine Learning for b-jet charge identification [48]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

2.20 Lattice Field Theories

2.20.1 Quantum Annealing

2.20.1.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [49]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

2.20.2 Quantum Simulations

2.20.2.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [50]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

2.20.2.2 Quantum Algorithms for Fermionic Quantum Field Theories [51]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

2.20.2.3 Digitization of scalar fields for quantum computing [52]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

2.20.2.4 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [53]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

2.20.2.5 General Methods for Digital Quantum Simulation of Gauge Theories [54]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

2.20.2.6 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [55]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

2.21 Lattice Gauge Theories

2.21.1 Quantum Annealing

2.21.1.1 SU(2) lattice gauge theory on a quantum annealer [56]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

2.21.2 Quantum Simulations

2.21.2.1 Simulating lattice gauge theories on a quantum computer [57]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

2.21.2.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [58]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

2.21.2.3 Towards Quantum Simulating QCD [59]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

2.21.2.4 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [60]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

2.21.2.5 Simulating Lattice Gauge Theories within Quantum Technologies [61]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

2.21.2.6 Toward scalable simulations of Lattice Gauge Theories on quantum computers [62]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

2.21.2.7 Quantum simulation of gauge theory via orbifold lattice [63]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

2.21.2.8 Lattice renormalization of quantum simulations [64]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

2.21.2.9 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [65]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

2.21.3 Variational Quantum Algorithms

2.21.3.1 SU(2) hadrons on a quantum computer via a variational approach [66]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

2.21.4 Crosslists

2.21.4.1 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [3]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

2.22 Lattice Quantum Chromodynamics

2.22.1 Quantum Simulations

2.22.1.1 Role of boundary conditions in quantum computations of scattering observables [67]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

2.23 Neutrinos

2.23.1 Variational Quantum Algorithms

2.23.1.1 Quantum convolutional neural networks for high energy physics data analysis [68]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.23.1.2 Hybrid Quantum-Classical Graph Convolutional Network [69]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

2.24 Quantum Chromodynamics

2.24.1 Quantum Entanglement

2.24.1.1 Symmetry from entanglement suppression [70]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
- **Published in Phys.Rev.D:** 01 October 2021

2.24.2 Quantum Simulations

2.24.2.1 Quantum simulation of quantum field theory in the light-front formulation [71]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

2.24.3 Crosslists

2.24.3.1 Partonic collinear structure by quantum computing [38]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

2.25 Quantum Field Theories

2.26 Spin-1/2 Bipartite Systems

2.26.1 Quantum Entanglement

2.26.1.1 Entanglement and quantum tomography with top quarks at the LHC [72]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

2.26.1.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [73]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
- **Posted on arXiv:** 23 February 2021
- **Published in Phys.Rev.Lett. 127 (2021) 16, 161801:** 15 October 2021

2.26.1.3 Quantum tops at the LHC: from entanglement to Bell inequalities [74]

- **Authors:** Claudio Severi, Cristian Degli Esposti Boschi, Fabio Maltoni, Maximiliano Sioli
- **Posted on arXiv:** 19 October 2021
- **Published in Eur.Phys.J.C:** 02 April 2022

2.26.1.4 Quantum SMEFT tomography: Top quark pair production at the LHC [23]

- **Authors:** Rafael Aoude, Eric Madge, Fabio Maltoni, Luca Mantani
- **Posted on arXiv:** 10 March 2022
- **Published in Phys.Rev.D:** 01 September 2022

2.26.1.5 Quantum information with top quarks in QCD [75]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

2.27 Statistical Analysis

2.27.1 Quantum Annealing

2.27.1.1 Unfolding measurement distributions via quantum annealing [76]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
- **Posted on arXiv:** 22 August 2019
- **Published in JHEP:** 22 November 2019

2.27.1.2 A Quantum Algorithm for Model-Independent Searches for New Physics [6]

- **Authors:** Konstantin T. Matchev, Prasanth Shyamsundar, Jordan Smolinsky
- **Posted on arXiv:** 04 March 2020
- **Published in LHEP:** 09 April 2023

2.28 Track Reconstruction

2.28.1 Quantum Algorithms - Grover's Search Algorithm

2.28.1.1 Quantum speedup for track reconstruction in particle accelerators [77]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

2.28.2 Quantum Annealing

2.28.2.1 A pattern recognition algorithm for quantum annealers [78]

- **Authors:** Frèdèric Bapst, Wahid Bhimji, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder
- **Posted on arXiv:** 21 February 2019
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2.28.2.2 Track clustering with a quantum annealer for primary vertex reconstruction at hadron colliders [79]

- **Authors:** Souvik Das, Andrew J. Wildridge, Andreas Jung
- **Posted on arXiv:** 21 March 2019

2.28.2.3 Charged particle tracking with quantum annealing optimization [80]

- **Authors:** Alexander Zlokapa, Abhishek Anand, Jean-Roch Vlimant, Javier M. Duarte, Joshua Job, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 12 August 2019
- **Published in Quantum Mach. Intell. 3, 27 (2021):** 02 November 2021

2.28.2.4 Quantum annealing algorithms for track pattern recognition [81]

- **Authors:** Masahiko Saito, Paolo Calafiura, Heather Gray, Wim Lavrijsen, Lucy Linder, Yasuyuki Okumura, Ryu Sawada, Alex Smith, Junichi Tanaka, Koji Terashi
- **Published in EPJ Web Conf.:** 2020

2.28.3 Quantum Neural Networks

2.28.3.1 Hybrid quantum classical graph neural networks for particle track reconstruction [82]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

2.28.4 Quantum Storage

2.28.4.1 Quantum Associative Memory in HEP Track Pattern Recognition [83]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

2.28.4.2 Particle track classification using quantum associative memory [84]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

2.28.5 Variational Quantum Algorithms

2.28.5.1 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [85]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

2.28.6 Crosslists

2.28.6.1 Quantum convolutional neural networks for high energy physics data analysis [68]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

2.28.6.2 Hybrid Quantum-Classical Graph Convolutional Network [69]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3 Quantum Information Science in Nuclear and Particle Physics

3.1 Reviews and Whitepapers

3.1.1 Reviews

3.1.1.1 Quantum Machine Learning in High Energy Physics [1]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.1.2 Whitepapers

3.1.2.1 Mechanical quantum sensing in the search for dark matter [2]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.1.2.2 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [3]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.2 Continuous Variable Quantum Computing

3.2.1 Detector Technology and Detector Simulation

3.2.1.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [14]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.2.2 Crosslists

3.2.2.1 Unsupervised event classification with graphs on classical and photonic quantum computers [4]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.3 Quantum Algorithms - Grover's Search Algorithm

3.3.1 Beyond the Standard Model - Dark Sector Searches

3.3.1.1 Application of a Quantum Search Algorithm to High- Energy Physics Data at the Large Hadron Collider [8]

- **Authors:** Anthony E. Armenakas, Oliver K. Baker
- **Posted on arXiv:** 01 October 2020

3.3.2 Event Classification

3.3.2.1 Implementation and analysis of quantum computing application to Higgs boson reconstruction at the large Hadron Collider [24]

- **Authors:** Anthony Alexiades Armenakas, Oliver K. Baker
- **Published in Sci.Rep.:** 24 November 2021

3.3.3 Event Generation

3.3.3.1 Quantum algorithm for Feynman loop integrals [32]

- **Authors:** Selomit Ramírez-Uribe, Andrès E. Rentería-Olivo, Germán Rodrigo, German F.R. Sborlini, Luiz Vale Silva
- **Posted on arXiv:** 18 May 2021
- **Published in JHEP:** 16 May 2022

3.3.4 Jet Reconstruction

3.3.4.1 Quantum Algorithms for Jet Clustering [41]

- **Authors:** Annie Y. Wei, Preksha Naik, Aram W. Harrow, Jesse Thaler
- **Posted on arXiv:** 23 August 2019
- **Published in Phys. Rev. D 101, 094015 (2020):** 15 May 2020

3.3.5 Track Reconstruction

3.3.5.1 Quantum speedup for track reconstruction in particle accelerators [77]

- **Authors:** Duarte Magano, Akshat Kumar, Mārtiņš Kālis, Andris Locāns, Adam Glos, Sagar Pratapsi, Gonçalo Quinta, Maksims Dimitrijevs, Aleksander Rivošs, Pedrame Bargassa, João Seixas, Andris Ambainis, Yasser Omar
- **Posted on arXiv:** 23 April 2021
- **Published in Physical Review D 105 (2022) 076012:** 01 April 2022

3.4 Quantum Annealing

3.4.1 Event Classification

3.4.1.1 Solving a Higgs optimization problem with quantum annealing for machine learning [25]

- **Authors:** Alex Mott, Joshua Job, Jean Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Published in Nature:** 18 October 2017

3.4.1.2 Quantum adiabatic machine learning by zooming into a region of the energy surface [26]

- **Authors:** Alexander Zlokapa, Alex Mott, Joshua Job, Jean-Roch Vlimant, Daniel Lidar, Maria Spiropulu
- **Posted on arXiv:** 13 August 2019
- **Published in Phys. Rev. A 102, 062405 (2020):** 05 December 2020

3.4.1.3 Quantum algorithm for the classification of supersymmetric top quark events [10]

- **Authors:** Pedrame Bargassa, Timothèe Cabos, Samuele Cavinato, Artur Cordeiro Oudot Choi, Timothèe Hessel
- **Posted on arXiv:** 31 May 2021
- **Published in Phys. Rev. D 104 (2021) 096004:** 01 November 2021

3.4.1.4 Completely quantum neural networks [27]

- **Authors:** Steve Abel, Juan C. Criado, Michael Spannowsky
- **Posted on arXiv:** 23 February 2022
- **Published in Phys.Rev.A:** 01 August 2022

3.4.2 Gravitation and Astrophysics

3.4.2.1 Restricted Boltzmann Machines for galaxy morphology classification with a quantum annealer [40]

- **Authors:** João Caldeira, Joshua Job, Steven H. Adachi, Brian Nord, Gabriel N. Perdue
- **Posted on arXiv:** 14 November 2019

3.4.3 Jet Reconstruction

3.4.3.1 Adiabatic quantum algorithm for multijet clustering in high energy physics [42]

- **Authors:** Diogo Pires, Yasser Omar, João Seixas
- **Posted on arXiv:** 28 December 2020
- **Published in Phys.Lett.B:** 13 June 2023

3.4.3.2 Leveraging Quantum Annealer to identify an Event-topology at High Energy Colliders [43]

- **Authors:** Minho Kim, Pyungwon Ko, Jae-hyeon Park, Myeonghun Park
- **Posted on arXiv:** 15 November 2021

3.4.4 Lattice Field Theories

3.4.4.1 A regression algorithm for accelerated lattice QCD that exploits sparse inference on the D-Wave quantum annealer [49]

- **Authors:** Nga T.T. Nguyen, Garrett T. Kenyon, Boram Yoon
- **Posted on arXiv:** 14 November 2019
- **Published in Sci Rep 10, 10915 (2020):** 02 July 2020

3.4.5 Lattice Gauge Theories

3.4.5.1 SU(2) lattice gauge theory on a quantum annealer [56]

- **Authors:** Sarmed A Rahman, Randy Lewis, Emanuele Mendicelli, Sarah Powell
- **Posted on arXiv:** 15 March 2021
- **Published in Phys. Rev. D 104, 034501 (2021):** 01 August 2021

3.4.6 Statistical Analysis

3.4.6.1 Unfolding measurement distributions via quantum annealing [76]

- **Authors:** Kyle Cormier, Riccardo Di Sipio, Peter Wittek
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3.4.6.2 A Quantum Algorithm for Model-Independent Searches for New Physics [6]

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3.4.8 Crosslists

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- **Posted on arXiv:** 23 August 2019
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3.4.8.2 Quantum Machine Learning in High Energy Physics [1]

- **Authors:** Wen Guan, Gabriel Perdue, Arthur Pesah, Maria Schuld, Koji Terashi, Sofia Vallecorsa, Jean-Roch Vlimant
- **Posted on arXiv:** 18 May 2020
- **Published in Mach.Learn.Sci.Tech.:** 31 March 2021

3.4.8.3 Particle track classification using quantum associative memory [84]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
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- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.5 Quantum Circuit Born Machines

3.6 Quantum Entanglement

3.6.1 Quantum Chromodynamics

3.6.1.1 Symmetry from entanglement suppression [70]

- **Authors:** Ian Low, Thomas Mehen
- **Posted on arXiv:** 21 April 2021
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3.6.2 Spin-1/2 Bipartite Systems

3.6.2.1 Entanglement and quantum tomography with top quarks at the LHC [72]

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3.6.2.2 Testing Bell Inequalities at the LHC with Top-Quark Pairs [73]

- **Authors:** M. Fabbrichesi, R. Floreanini, G. Panizzo
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3.6.2.5 Quantum information with top quarks in QCD [75]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 10 March 2022
- **Published in Quantum:** 29 September 2022

3.7 Quantum Generative Adversarial Networks

3.7.1 Detector Technology and Detector Simulation

3.7.1.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [15]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.7.2 Event Generation

3.7.2.1 Style-based quantum generative adversarial networks for Monte Carlo events [33]

- **Authors:** Carlos Bravo-Prieto, Julien Baglio, Marco Cè, Anthony Francis, Dorota M. Grabowska, Stefano Carrazza
- **Posted on arXiv:** 13 October 2021
- **Published in Quantum 6, 777 (2022):** 17 August 2022

3.7.2.2 Quantum integration of elementary particle processes [34]

- **Authors:** Gabriele Agliardi, Michele Grossi, Mathieu Pellen, Enrico Prati
- **Posted on arXiv:** 05 January 2022
- **Published in Phys.Lett.B:** 07 June 2022

3.7.3 Crosslists

3.7.3.1 Quantum Generative Adversarial Networks in a Continuous-Variable Architecture to Simulate High Energy Physics Detectors [14]

- **Authors:** Su Yeon Chang, Sofia Vallecorsa, Elías F. Combarro, Federico Carminati
- **Posted on arXiv:** 26 January 2021

3.8 Quantum Kernel Methods

3.8.1 Event Classification

3.8.1.1 Quantum Support Vector Machines for Continuum Suppression in B Meson Decays [28]

- **Authors:** Jamie Heredge, Charles Hill, Lloyd Hollenberg, Martin Sevior
- **Posted on arXiv:** 22 March 2021
- **Published in Comput.Softw.Big Sci.:** 30 November 2021

3.8.1.2 Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [29]

- **Authors:** Sau Lan Wu, Shaojun Sun, Wen Guan, Chen Zhou, Jay Chan, Chi Lung Cheng, Tuan Pham, Yan Qian, Alex Zeng Wang, Rui Zhang, Miron Livny, Jennifer Glick, Panagiotis Kl. Barkoutsos, Stefan Woerner, Ivano Tavernelli, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 11 April 2021
- **Published in Phys. Rev. Research 3, 033221 (2021):** 08 September 2021

3.8.1.3 Higgs analysis with quantum classifiers [30]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.9 Quantum Inspired Algorithms

3.10 Quantum k-Means

3.10.1 Anomaly Detection

3.10.1.1 Unsupervised event classification with graphs on classical and photonic quantum computers [4]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 05 March 2021
- **Published in J. High Energ. Phys. 2021, 170:** 31 August 2021

3.10.2 Jet Reconstruction

3.10.2.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [44]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.11 Quantum Neural Networks

3.11.1 Track Reconstruction

3.11.1.1 Hybrid quantum classical graph neural networks for particle track reconstruction [82]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

3.11.2 Crosslists

3.11.2.1 Quantum convolutional neural networks for high energy physics data analysis [68]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.12 Quantum Sensors - Levitated Sensors

3.12.1 Dark Matter - Large Composite States

3.12.1.1 Search for composite dark matter with optically levitated sensors [18]

- **Authors:** Fernando Monteiro, Gadi Afek, Daniel Carney, Gordan Krnjaic, Jiaxiang Wang, David C. Moore
- **Posted on arXiv:** 23 July 2020
- **Published in Phys. Rev. Lett. 125, 181102 (2020):** 29 October 2020

3.12.2 Dark Matter - Low Mass Dark Matter

3.12.2.1 Coherent Scattering of Low Mass Dark Matter from Optically Trapped Sensors [19]

- **Authors:** Gadi Afek, Daniel Carney, David C. Moore
- **Posted on arXiv:** 05 November 2021
- **Published in Phys.Rev.Lett.:** 09 March 2022

3.13 Quantum Sensors - Mechanical Detectors

3.13.1 Crosslists

3.13.1.1 Mechanical quantum sensing in the search for dark matter [2]

- **Authors:** D. Carney, G. Krnjaic, D.C. Moore, C.A. Regal, G. Afek, S. Bhave, B. Brubaker, T. Corbitt, J. Cripe, N. Crisosto, A. Geraci, S. Ghosh, J.G.E. Harris, A. Hook, E.W. Kolb, J. Kunjummen, R.F. Lang, T. Li, T. Lin, Z. Liu, J. Lykken, L. Magrini, J. Manley, N. Matsumoto, A. Monte, F. Monteiro, T. Purdy, C.J. Riedel, R. Singh, S. Singh, K. Sinha, J.M. Taylor, J. Qin, D.J. Wilson, Y. Zhao
- **Posted on arXiv:** 13 August 2020
- **Published in Quantum Sci. Technol. 6 024002, 2021 (invited):** 18 January 2021

3.14 Quantum Sensors - Nitrogen-Vacancy Centers

3.14.1 Dark Matter - Axions and Axion-like Particles

3.14.1.1 Searching for an exotic spin-dependent interaction with a single electron-spin quantum sensor [11]

- **Authors:** Xing Rong, Mengqi Wang, Jianpei Geng, Xi Qin, Maosen Guo, Man Jiao, Yijin Xie, Pengfei Wang, Pu Huang, Fazhan Shi, Yi-Fu Cai, Chongwen Zou, Jiangfeng Du

- **Posted on arXiv:** 12 June 2017
- **Published in Nature Commun.:** 21 February 2018

3.14.1.2 Ultrasensitive optomechanical detection of an axion-mediated force based on a sharp peak emerging in probe absorption spectrum [12]

- **Authors:** Lei Chen, Jian Liu, Kadi Zhu
- **Posted on arXiv:** 28 November 2019
- **Published in Opt.Commun.:** 15 January 2021

3.14.2 Dark Matter - Exotic Dark Matter

3.14.2.1 Proposal for the search for new spin interactions at the micrometer scale using diamond quantum sensors [16]

- **Authors:** P.-H. Chu, N. Ristoff, J. Smits, N. Jackson, Y.J. Kim, I. Savukov, V.M. Acosta
- **Posted on arXiv:** 29 December 2021
- **Published in Physical Review Research 4, 023162 (2022):** 31 May 2022

3.15 Quantum Sensors - Ultracold Atoms in Optical Lattices

3.16 Quantum Sensors - Superconducting Transmon Qubits

3.16.1 Dark Matter - Axions and Axion-like Particles

3.16.1.1 Accelerating dark-matter axion searches with quantum measurement technology [13]

- **Authors:** Huaixiu Zheng, Matti Silveri, R.T. Brierley, S.M. Girvin, K.W. Lehnert
- **Posted on arXiv:** 08 July 2016

3.16.2 Dark Matter - Hidden/Dark Photons

3.16.2.1 Searching for Dark Matter with a Superconducting Qubit [17]

- **Authors:** Akash V. Dixit, Srivatsan Chakram, Kevin He, Ankur Agrawal, Ravi K. Naik, David I. Schuster, Aaron Chou
- **Posted on arXiv:** 28 August 2020
- **Published in Phys. Rev. Lett. 126, 141302 (2021):** 09 April 2021

3.17 Quantum Sensors - Rydberg Atoms

3.18 Quantum Sensors - Trapped Ions

3.18.1 Dark Matter - Millicharged Particles

3.18.1.1 Trapped Electrons and Ions as Particle Detectors [20]

- **Authors:** Daniel Carney, Hartmut Häffner, David C. Moore, Jacob M. Taylor
- **Posted on arXiv:** 12 April 2021
- **Published in Phys Rev Lett, Vol. 127, No. 6, 2021. "Editor's suggestion":** 05 August 2021

3.18.1.2 Millicharged Dark Matter Detection with Ion Traps [21]

- **Authors:** Dmitry Budker, Peter W. Graham, Harikrishnan Ramani, Ferdinand Schmidt-Kaler, Christian Smorra, Stefan Ulmer
- **Posted on arXiv:** 11 August 2021
- **Published in PRX Quantum:** 01 February 2022

3.19 Quantum Simulations

3.19.1 Effective Field Theories

3.19.1.1 Simulating Collider Physics on Quantum Computers Using Effective Field Theories [22]

- **Authors:** Christian W. Bauer, Marat Freytsis, Benjamin Nachman
- **Posted on arXiv:** 09 February 2021
- **Published in Phys.Rev.Lett.:** 18 November 2021

3.19.2 Event Generation

3.19.2.1 Quantum Algorithm for High Energy Physics Simulations [35]

- **Authors:** Christian W. Bauer, Wibe A. de Jong, Benjamin Nachman, Davide Provasoli
- **Posted on arXiv:** 05 April 2019
- **Published in Phys. Rev. Lett. 126, 062001 (2021):** 11 February 2021

3.19.2.2 Towards a quantum computing algorithm for helicity amplitudes and parton showers [36]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 13 October 2020
- **Published in Phys. Rev. D 103, 076020 (2021):** 27 April 2021

3.19.3 Lattice Field Theories

3.19.3.1 Quantum Computation of Scattering in Scalar Quantum Field Theories [50]

- **Authors:** Stephen P. Jordan, Keith S.M. Lee, John Preskill
- **Posted on arXiv:** December 2011

3.19.3.2 Quantum Algorithms for Fermionic Quantum Field Theories [51]

- **Authors:** Stephen P. Jordan, Keith S. M. Lee, John Preskill
- **Posted on arXiv:** 28 April 2014

3.19.3.3 Digitization of scalar fields for quantum computing [52]

- **Authors:** Natalie Klco, Martin J. Savage
- **Posted on arXiv:** 30 August 2018
- **Published in Phys. Rev. A 99, 052335 (2019):** 23 May 2019

3.19.3.4 Scalar Quantum Field Theories as a Benchmark for Near-Term Quantum Computers [53]

- **Authors:** Kubra Yeter-Aydeniz, Eugene F. Dumitrescu, Alex J. McCaskey, Ryan S. Bennink, Raphael C. Pooser, George Siopsis
- **Posted on arXiv:** 29 November 2018
- **Published in Phys. Rev. A 99, 032306 (2019):** 04 March 2019

3.19.3.5 General Methods for Digital Quantum Simulation of Gauge Theories [54]

- **Authors:** Henry Lamm, Scott Lawrence, Yukari Yamauchi
- **Posted on arXiv:** 19 March 2019
- **Published in Phys. Rev. D 100, 034518 (2019):** 29 August 2019

3.19.3.6 SU(2) non-Abelian gauge field theory in one dimension on digital quantum computers [55]

- **Authors:** Natalie Klco, Jesse R. Stryker, Martin J. Savage
- **Posted on arXiv:** 19 August 2019
- **Published in Phys. Rev. D 101, 074512 (2020):** 22 April 2020

3.19.4 Lattice Gauge Theories

3.19.4.1 Simulating lattice gauge theories on a quantum computer [57]

- **Authors:** Tim Byrnes, Yoshihisa Yamamoto
- **Posted on arXiv:** October 2005
- **Published in Phys.Rev.A:** 2006

3.19.4.2 Ultracold Quantum Gases and Lattice Systems: Quantum Simulation of Lattice Gauge Theories [58]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 07 May 2013
- **Published in Annalen Phys.:** 2013

3.19.4.3 Towards Quantum Simulating QCD [59]

- **Authors:** Uwe-Jens Wiese
- **Posted on arXiv:** 25 September 2014
- **Published in Nucl.Phys.A:** 28 September 2014

3.19.4.4 Real-time dynamics of lattice gauge theories with a few-qubit quantum computer [60]

- **Authors:** E.A. Martinez, C.A. Muschik, P. Schindler, D. Nigg, A. Erhard, M. Heyl, P. Hauke, M. Dalmonte, T. Monz, P. Zoller, R. Blatt
- **Posted on arXiv:** 15 May 2016
- **Published in Nature:** 22 June 2016

3.19.4.5 Simulating Lattice Gauge Theories within Quantum Technologies [61]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.19.4.6 Toward scalable simulations of Lattice Gauge Theories on quantum computers [62]

- **Authors:** Simon V. Mathis, Guglielmo Mazzola, Ivano Tavernelli
- **Posted on arXiv:** 20 May 2020
- **Published in Phys. Rev. D 102, 094501 (2020):** 04 November 2020

3.19.4.7 Quantum simulation of gauge theory via orbifold lattice [63]

- **Authors:** Alexander J. Buser, Hrant Gharibyan, Masanori Hanada, Masazumi Honda, Junyu Liu
- **Posted on arXiv:** 12 November 2020
- **Published in JHEP 2109 (2021) 034:** 06 September 2021

3.19.4.8 Lattice renormalization of quantum simulations [64]

- **Authors:** Marcela Carena, Henry Lamm, Ying-Ying Li, Wanqiang Liu
- **Posted on arXiv:** 02 July 2021
- **Published in Phys.Rev.D:** 01 November 2021

3.19.4.9 Efficient representation for simulating U(1) gauge theories on digital quantum computers at all values of the coupling [65]

- **Authors:** Christian W. Bauer, Dorota M. Grabowska
- **Posted on arXiv:** 15 November 2021
- **Published in Phys.Rev.D:** 01 February 2023

3.19.5 Lattice Quantum Chromodynamics

3.19.5.1 Role of boundary conditions in quantum computations of scattering observables [67]

- **Authors:** Raúl A. Briceño, Juan V. Guerrero, Maxwell T. Hansen, Alexandru M. Sturzu
- **Posted on arXiv:** 01 July 2020
- **Published in Phys. Rev. D 103, 014506 (2021):** 07 January 2021

3.19.6 Quantum Chromodynamics

3.19.6.1 Quantum simulation of quantum field theory in the light-front formulation [71]

- **Authors:** Michael Kreshchuk, William M. Kirby, Gary Goldstein, Hugo Beauchemin, Peter J. Love
- **Posted on arXiv:** 10 February 2020
- **Published in Phys. Rev. A 105, 032418 (2022):** 08 March 2022

3.20 Quantum Storage

3.20.1 Track Reconstruction

3.20.1.1 Quantum Associative Memory in HEP Track Pattern Recognition [83]

- **Authors:** Illya Shapoval, Paolo Calafiura
- **Posted on arXiv:** 29 January 2019
- **Published in EPJ Web of Conferences 214 (2019) 01012:** 2019

3.20.1.2 Particle track classification using quantum associative memory [84]

- **Authors:** Gregory Quiroz, Lauren Ice, Andrea Delgado, Travis S. Humble
- **Posted on arXiv:** 23 November 2020
- **Published in Nucl.Instrum.Meth.A:** 15 June 2021

3.20.2 Crosslists

3.20.2.1 A Digital Quantum Algorithm for Jet Clustering in High-Energy Physics [44]

- **Authors:** Diogo Pires, Pedrame Bargassa, João Seixas, Yasser Omar
- **Posted on arXiv:** 11 January 2021

3.21 Quantum Tomography

3.21.1 Crosslists

3.21.1.1 Entanglement and quantum tomography with top quarks at the LHC [72]

- **Authors:** Yoav Afik, Juan Ramón Muñoz de Nova
- **Posted on arXiv:** 04 March 2020
- **Published in Eur. Phys. J. Plus (2021) 136:907:** 03 September 2021

3.22 Quantum Unsupervised Clustering Algorithms

3.23 Quantum Walks

3.23.1 Event Generation

3.23.1.1 Quantum walk approach to simulating parton showers [37]

- **Authors:** Khadeejah Bepari, Sarah Malik, Michael Spannowsky, Simon Williams
- **Posted on arXiv:** 28 September 2021
- **Published in Phys. Rev. D 106 (2022) 056002:** 01 September 2022

3.24 Tensor Networks

3.24.1 Jet Reconstruction

3.24.1.1 Quantum-inspired machine learning on high-energy physics data [45]

- **Authors:** Timo Felser, Marco Trenti, Lorenzo Sestini, Alessio Gianelle, Davide Zuliani, Donatella Lucchesi, Simone Montangero
- **Posted on arXiv:** 28 April 2020
- **Published in npj Quantum Inf.:** 15 July 2021

3.24.1.2 Quantum-inspired event reconstruction with Tensor Networks: Matrix Product States [46]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 15 June 2021
- **Published in JHEP 08 (2021) 112:** 23 August 2021

3.24.1.3 Classical versus quantum: Comparing tensor-network-based quantum circuits on Large Hadron Collider data [47]

- **Authors:** Jack Y. Araz, Michael Spannowsky
- **Posted on arXiv:** 21 February 2022
- **Published in Phys. Rev. A 106 (Dec, 2022) 062423:** 19 December 2022

3.24.2 Crosslists

3.24.2.1 Simulating Lattice Gauge Theories within Quantum Technologies [61]

- **Authors:** M.C. Bañuls, R. Blatt, J. Catani, A. Celi, J.I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C.A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, P. Zoller
- **Posted on arXiv:** 31 October 2019
- **Published in Eur.Phys.J.D:** 04 August 2020

3.24.2.2 Tensor networks for High Energy Physics: contribution to Snowmass 2021 [3]

- **Authors:** Yannick Meurice, James C. Osborn, Ryo Sakai, Judah Unmuth-Yockey, Simon Catterall, Rolando D. Somma
- **Posted on arXiv:** 09 March 2022

3.25 Variational Quantum Algorithms

3.25.1 Anomaly Detection

3.25.1.1 Anomaly detection in high-energy physics using a quantum autoencoder [5]

- **Authors:** Vishal S. Ngairangbam, Michael Spannowsky, Michihisa Takeuchi
- **Posted on arXiv:** 09 December 2021
- **Published in Phys. Rev. D 105, 095004, 2022:** 01 May 2022

3.25.2 Event Classification

3.25.2.1 Event Classification with Quantum Machine Learning in High-Energy Physics [9]

- **Authors:** Koji Terashi, Michiru Kaneda, Tomoe Kishimoto, Masahiko Saito, Ryu Sawada, Junichi Tanaka
- **Posted on arXiv:** 23 February 2020
- **Published in Comput. Softw. Big Sci. 5, 2 (2021):** 03 January 2021

3.25.2.2 Quantum Machine Learning for Particle Physics using a Variational Quantum Classifier [7]

- **Authors:** Andrew Blance, Michael Spannowsky
- **Posted on arXiv:** 14 October 2020
- **Published in JHEP:** 2021

3.25.2.3 Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [31]

- **Authors:** Sau Lan Wu, Jay Chan, Wen Guan, Shaojun Sun, Alex Wang, Chen Zhou, Miron Livny, Federico Carminati, Alberto Di Meglio, Andy C.Y. Li, Joseph Lykken, Panagiotis Spentzouris, Samuel Yen-Chi Chen, Shinjae Yoo, Tzu-Chieh Wei
- **Posted on arXiv:** 21 December 2020
- **Published in J.Phys.G:** 26 October 2021

3.25.3 Event Generation

3.25.3.1 Partonic collinear structure by quantum computing [38]

- **Authors:** Tianyin Li, Xingyu Guo, Wai Kin Lai, Xiaohui Liu, Enke Wang, Hongxi Xing, Dan-Bo Zhang, Shi-Liang Zhu
- **Posted on arXiv:** 07 June 2021
- **Published in Phys.Rev.D:** 01 June 2022

3.25.3.2 Unsupervised quantum circuit learning in high energy physics [39]

- **Authors:** Andrea Delgado, Kathleen E. Hamilton
- **Posted on arXiv:** 07 March 2022
- **Published in Phys.Rev.D:** 01 November 2022

3.25.4 Jet Reconstruction

3.25.4.1 Quantum Machine Learning for b-jet charge identification [48]

- **Authors:** Alessio Gianelle, Patrick Koppenburg, Donatella Lucchesi, Davide Nicotra, Eduardo Rodrigues, Lorenzo Sestini, Jacco de Vries, Davide Zuliani
- **Posted on arXiv:** 28 February 2022
- **Published in JHEP:** 01 August 2022

3.25.5 Lattice Gauge Theories

3.25.5.1 SU(2) hadrons on a quantum computer via a variational approach [66]

- **Authors:** Yasar Y. Atas, Jinglei Zhang, Randy Lewis, Amin Jahanpour, Jan F. Haase, Christine A. Muschik
- **Posted on arXiv:** 17 February 2021
- **Published in Nature Communications 2021:** 11 November 2021

3.25.6 Neutrinos

3.25.6.1 Quantum convolutional neural networks for high energy physics data analysis [68]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 22 December 2020
- **Published in Phys.Rev.Res.:** 28 March 2022

3.25.6.2 Hybrid Quantum-Classical Graph Convolutional Network [69]

- **Authors:** Samuel Yen-Chi Chen, Tzu-Chieh Wei, Chao Zhang, Haiwang Yu, Shinjae Yoo
- **Posted on arXiv:** 15 January 2021

3.25.7 Track Reconstruction

3.25.7.1 Studying quantum algorithms for particle track reconstruction in the LUXE experiment [85]

- **Authors:** Lena Funcke, Tobias Hartung, Beate Heinemann, Karl Jansen, Annabel Kropf, Stefan Kühn, Federico Meloni, David Spataro, Cenk Tüysüz, Yee Chinn Yap
- **Posted on arXiv:** 14 February 2022
- **Published in J.Phys.Conf.Ser.:** 2023

3.25.8 Crosslists

3.25.8.1 Dual-Parameterized Quantum Circuit GAN Model in High Energy Physics [15]

- **Authors:** Su Yeon Chang, Steven Herbert, Sofia Vallecorsa, Elías F. Combarro, Ross Duncan
- **Posted on arXiv:** 29 March 2021
- **Published in EPJ Web Conf.:** 2021

3.25.8.2 Higgs analysis with quantum classifiers [30]

- **Authors:** Vasileios Belis, Samuel González-Castillo, Christina Reissel, Sofia Vallecorsa, Elías F. Combarro, Günther Dissertori, Florentin Reiter
- **Posted on arXiv:** 15 April 2021
- **Published in EPJ Web Conf.:** 2021

3.25.8.3 Hybrid quantum classical graph neural networks for particle track reconstruction [82]

- **Authors:** Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster
- **Posted on arXiv:** 26 September 2021
- **Published in Quantum Machine Intelligence:** 01 December 2021

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